Lecture 8

Tangent spaces, Lie groups and homogeneous spaces

Objectives. *A tangent vector records the initial velocity of a curve without choosing an ambient Euclidean space. Coordinate changes relate its components by their Jacobian matrices. We use these vectors to differentiate maps and describe the infinitesimal directions of matrix groups. A transitive smooth action of a Lie group presents a manifold as a quotient by the subgroup fixing a point. The sphere and the qubit state space give two concrete quotient descriptions.*

By the end of this lecture you will be able to:

- Define tangent vectors by curves and transform their components.
- Compute differentials and tangent spaces to regular level sets.
- Identify the tangent space of $\mathrm{SO}(3)$ at its identity.
- Determine stabilisers and justify the sphere quotient descriptions.

8.1 Tangent vectors as velocities

Let p be a point of a smooth n-manifold M. A *smooth curve through* p is a smooth map $\gamma\colon (-\varepsilon,\varepsilon) \to M$ with $\gamma(0) = p$, where $\varepsilon > 0$. In a chart $x = (x^1, \dots, x^n)$ near p, its coordinate velocity is $(x \circ \gamma)'(0) \in \mathbb{R}^n$. Two such curves are equivalent if these velocities agree. A *tangent vector* at p is an equivalence class of curves. The set of these classes is the *tangent space*, written T_pM. The drawing shows curves with different second-order behaviour but the same initial velocity in one chart.

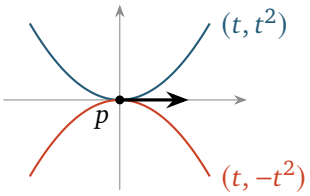


Figure 8.1: The coordinate curves (t, t^2) and $(t, -t^2)$ have the same velocity $(1, 0)$ at the marked point. Their common tangent vector does not record their bending.

Theorem 8.1 (Velocity classes and components)**.** *Curve equivalence is independent of the chart. A chart identifies T_pM with $\mathbb{R}^n$, giving it an n-dimensional vector-space structure independent of that chart. Under a coordinate change $y = y(x)$, components transform by*

$$v^a_y = \frac{\partial y^a}{\partial x^i}(p)\, v^i_x.$$

Proof. The chain rule gives $(y \circ \gamma)'(0) = D(y \circ x^{-1})_{x(p)}(x \circ \gamma)'(0)$. Thus equal velocities in x give equal velocities in y. The reverse coordinate change proves the converse. This also gives the displayed component formula.

The velocity map $T_pM \to \mathbb{R}^n$ is injective by the definition of equivalence. For any $v \in \mathbb{R}^n$, the curve $\gamma(t) = x^{-1}(x(p) + tv)$ is defined for sufficiently small $|t|$ because the chart image is open. It has velocity v, proving surjectivity. Transport addition and scalar multiplication through this bijection. A change of chart acts by an invertible linear Jacobian, so it preserves these operations. This proves that the vector-space structure is independent of the chart. ∎

The basis vector with coordinate components equal to the ith unit column is written $\partial_i|_p = \partial/\partial x^i|_p$. Thus $v = v^i \partial_i|_p$, with the upper/lower index convention of Lecture 7. The symbols $\partial_i|_p$ denote tangent vectors, not points of M.

A tangent vector differentiates smooth functions near p by

$$v(f) = \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} f(\gamma(t)) = v^i \frac{\partial (f \circ x^{-1})}{\partial x^i}(x(p)).$$

The right side shows independence of the representative curve. The ordinary product rule gives $v(fg) = v(f)g(p) + f(p)v(g)$. Such a real linear rule on functions is called a *point derivation*. Only the function near p matters. Formally a *germ* at p is an equivalence class of functions defined near p, with two functions equivalent when they agree on some smaller neighbourhood. The product rule makes sense on germs. Conversely, all point derivations on smooth germs have the displayed coordinate form. This comparison is cited from Tu, §8, Proposition 8.9; its derivation formulation is developed on Sheet 8.

8.2 Differentials and regular constraints

For a smooth map $f\colon M \to N$, its *differential* at p is

$$\mathrm{d}f_p\colon T_pM \longrightarrow T_{f(p)}N, \qquad \mathrm{d}f_p[\gamma] = [f \circ \gamma].$$

In coordinates the image velocity is $D\widehat{f}_{x(p)}v_x$, by the chain rule. This depends only on the velocity class and is linear. It therefore defines the stated linear map. For $g\colon N \to P$, composition on representatives gives the intrinsic chain rule

$$\mathrm{d}(g \circ f)_p = \mathrm{d}g_{f(p)} \circ \mathrm{d}f_p. \tag{8.1}$$

The identity map has identity differential. Applying this formula to a diffeomorphism and its inverse shows that their differentials are inverse linear maps.

For example, if $f\colon \mathbb{R}^2 \to \mathbb{R}^3$ is given by $f(s,t) = (s, t, s^2 + st)$, then

$$\mathrm{d}f_{(s,t)}(u,v) = (u, v, (2s+t)u + sv).$$

Its first two components recover (u,v), so the differential is injective. Recall from Lecture 7 that an injective differential defines an *immersion*. An onto differential defines a *submersion*. An immersion that is a homeomorphism onto its image is an *embedding*.

Proposition 8.2 (Tangent space of a level set · *proved given the implicit function theorem*). *Let $S = f^{-1}(c) \subset W \subseteq \mathbb{R}^N$ be a nonempty regular level set of a smooth map $f\colon W \to \mathbb{R}^k$, with W open. The inclusion identifies T_pS with $\ker Df_p$ for every $p \in S$.*

Proof. For a curve in S, differentiating $f(\gamma(t)) = c$ gives $Df_p\gamma'(0) = 0$. In the graph chart of Theorem 7.3, the inclusion has the form $s \mapsto (s, h(s))$. Its differential sends u to $(u, Dh\,u)$, so it is injective. Its image has dimension $N - k$. The kernel of the onto map Df_p has that same dimension by rank–nullity. The inclusion already proved is therefore equality. ∎

For S^2, the constraint $|p|^2 = 1$ gives $T_pS^2 = \{v \in \mathbb{R}^3 : p \cdot v = 0\}$, since the derivative is $v \mapsto 2p \cdot v$. Thus the intrinsic velocity classes agree with the familiar tangent plane in this embedded example.

8.3 The tangent bundle

The *tangent bundle* is the disjoint union of all tangent spaces:

$$TM = \{(p, v) : p \in M,\ v \in T_pM\}.$$

The projection $\pi_{TM} : TM \to M$ is given by $\pi_{TM}(p, v) = p$. Its *fibre* over p is the inverse image $\pi_{TM}^{-1}(p)$, identified with T_pM. Vectors at different base points belong to different fibres. They cannot be added without a further identification.

Proposition 8.3 (Manifold structure on the tangent bundle · *sketched*)**.** *If M is a smooth n-manifold, then TM has a natural smooth structure of dimension $2n$. The projection is a smooth submersion.*

We sketch its chart construction. A chart x on $U \subseteq M$ gives coordinates $(x(p), v_x)$ on $\pi_{TM}^{-1}(U)$ with image $x(U) \times \mathbb{R}^n$. If y is another chart, the overlap map is

$$(x, v_x) \longmapsto \big(y(x), Dy_x v_x\big).$$

It is smooth and has the inverse obtained by the inverse coordinate change. The base and velocity coordinates give $2n$ independent real coordinates. They also specify the topology. Distinct base points can be separated downstairs. Distinct vectors over the same point can be separated in a common bundle chart. A countable atlas downstairs gives a countable basis in these product coordinates. This gives Hausdorffness and second countability. In each chart the projection is $(x, v) \mapsto x$, whose derivative is onto. The detailed chart verification is in Tu, §12, and Sheet 8.

For the circle, write $p = (\cos\theta, \sin\theta)$. Every tangent vector has the unique form $v = r(-\sin\theta, \cos\theta)$ with $r \in \mathbb{R}$. These formulas identify TS^1 with $S^1 \times \mathbb{R}$: the angle is periodic but the velocity coefficient is unrestricted. The figure displays this cylinder and its projection.

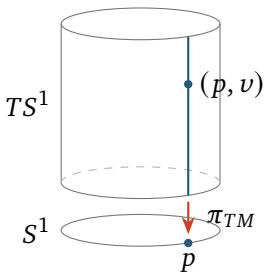


Figure 8.2: A fibre of TS^1 is a line of velocity coefficients. Projection forgets the coefficient and retains the point of the circle.

8.4 Lie groups and tangent matrices

A *Lie group* is a smooth manifold G that is also a group, with smooth multiplication $G\times G \to G$ and smooth inversion $G \to G$. Write e for its identity. For $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}\}$, the *general linear group* is

$$\mathrm{GL}(n, \mathbb{K}) = \{A \in \mathbb{K}^{n\times n} : \det A \neq 0\}.$$

The *special orthogonal group* is

$$\mathrm{SO}(n) = \{A \in \mathrm{O}(n) : \det A = 1\}.$$

A *matrix subgroup* is a subgroup of a general linear group. For complex matrices, the ambient space is regarded as a real manifold.

Proposition 8.4 (Regular matrix groups · *proved given the regular level set theorem*)**.** *A matrix subgroup that is a regular level set has the induced structure of a Lie group. The tangent space $T_I\mathrm{SO}(3)$ is the vector space $\mathfrak{so}(3) = \{A \in \mathbb{R}^{3\times 3} : A^T = -A\}$.*

Proof. The regular level set theorem gives an embedded manifold. Matrix multiplication is polynomial in real entries. Matrix inversion is smooth on invertible matrices, by the adjugate formula divided by the nonzero determinant. These maps preserve the subgroup. In adapted submanifold charts, their restrictions retain the first manifold-coordinate components of their ambient expressions, so the restrictions are smooth. They are the group's multiplication and inversion, as required.

The orthogonal constraint and its surjective derivative were computed in Lecture 7. The determinant-one part is open in $\mathrm{O}(3)$, since its only determinants are $+1$ and -1. It therefore has the same tangent space at I. By Proposition 8.2, that space is the kernel of $A \mapsto A^T + A$. This is precisely the antisymmetric matrices. ∎

Such a matrix has the unique form

$$A(a, b, c) = \begin{pmatrix} 0 & -c & b \\ c & 0 & -a \\ -b & a & 0 \end{pmatrix}, \qquad (a, b, c) \in \mathbb{R}^3.$$

Thus $\mathfrak{so}(3)$ has basis $A(1, 0, 0)$, $A(0, 1, 0)$ and $A(0, 0, 1)$. These are infinitesimal rotations in the fixed Cartesian basis.

A *Lie algebra* is a real vector space with an antisymmetric bilinear bracket satisfying the Jacobi identity $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$. For a Lie group embedded as a matrix subgroup, the tangent space $\mathfrak{g} = T_I G$ carries the matrix commutator $[A, B] = AB - BA$. To show closure, choose curves $a(s)$ and $b(t)$ in the group with initial value I and velocities A and B. Differentiating $a(s)b(t)a(s)^{-1}$ first in t gives $a(s)Ba(s)^{-1} \in \mathfrak{g}$. Differentiating in s at zero gives $AB - BA$, since $(a^{-1})'(0) = -A$, obtained by differentiating $a(s)a(s)^{-1} = I$. This derivative stays in the finite-dimensional vector subspace $\mathfrak{g}$. Bilinearity and antisymmetry follow from the formula. One nested commutator expands as

$$[A, [B, C]] = ABC - ACB - BCA + CBA.$$

In the sum with its two cyclic versions, each product occurs once with each sign. This proves Jacobi. For an abstract Lie group, Lecture 9 constructs the bracket on T_eG using vector fields.

8.5 Actions and homogeneous spaces

A *smooth left action* of a Lie group on M is a smooth map $G \times M \to M$, written $(g, p) \mapsto g \cdot p$, satisfying $e \cdot p = p$ and $(gh) \cdot p = g \cdot (h \cdot p)$. The *orbit* of p is $\{g \cdot p : g \in G\}$. The *orbit map* at p is the map $G \to M$ given by $g \mapsto g \cdot p$. The action is *transitive* when this orbit is all of M. A manifold with a transitive smooth action is a *homogeneous space*. The *stabiliser* (or isotropy subgroup) of p is

$$H = \{g \in G : g \cdot p = p\}.$$

It is a subgroup by the action laws. It is closed, being the inverse image of the closed singleton $\{p\}$ under the continuous orbit map.

For a subgroup H, the *left coset* gH is $\{gh : h \in H\}$. The set of left cosets, with the quotient topology, is written G/H. Two elements have the same coset exactly when $g_1^{-1} g_2 \in H$. An *embedded Lie subgroup* is a subgroup with an embedded smooth manifold structure and the restricted smooth group operations.

> **Theorem 8.5** (Closed subgroups and homogeneous spaces · *cited*)**.** *A closed subgroup H of a Lie group is an embedded Lie subgroup. The quotient G/H has a smooth manifold structure of dimension $\dim G - \dim H$ for which the quotient map is a smooth submersion. For a transitive smooth action on M with stabiliser H at p, the map $gH \mapsto g \cdot p$ is a diffeomorphism $G/H \to M$.*

These are Lee, second edition, Theorems 20.12, 21.17 and 21.18. The action identifies cosets with points: if $g_1 \cdot p = g_2 \cdot p$, then $g_1^{-1} g_2 \in H$, and the converse follows from the action law. Transitivity gives surjectivity, completing the bijection. The smooth-manifold and diffeomorphism assertions require the cited theorems.

Closedness cannot be omitted. The irrational line of Lecture 7 is a subgroup H of T^2, since adding its parameters multiplies its circle components. It is dense. Indeed, arbitrarily small nonzero fractional parts of integer multiples of α exist by the pigeonhole argument there, allowing either sign of the integer. If such a fractional part is $0 < \delta < \varepsilon$, the multiples $0, \delta, 2\delta, \ldots$ up to the last below 1 approach every point of $[0, 1)$ within ε. Thus integer multiples of α are dense modulo one. Fixing the first circle coordinate of the line leaves all these choices for the second, proving density in the torus. The subgroup is proper: over first coordinate 1, only the countably many points with second coordinate $e^{2\pi i \alpha m}$, $m \in \mathbb{Z}$, occur. Hence H is not closed. If T^2/H were Hausdorff, its singleton identity coset would be closed. Its inverse image H would then be closed, a contradiction. This quotient is not a manifold of our definition.

8.6 The sphere and the Hopf map

The standard action of $\mathrm{SO}(n+1)$ on S^n is transitive. Given a unit vector, extend it to an oriented orthonormal basis by Gram–Schmidt, recalled from *Linear Algebra for Physics*. Use it as the last column of an orthogonal matrix, completing the other columns with positive orientation. That matrix sends the last coordinate unit vector to the chosen point. The stabiliser consists of matrices with a block in $\mathrm{SO}(n)$ on its orthogonal complement and a last diagonal entry 1. The group $\mathrm{O}(n+1)$ is a regular level set by Lecture 7, so it is a Lie group by Proposition 8.4. Its determinant-one part is open and hence is also a Lie group. The sphere action is the restriction of the polynomial map $(g, p) \mapsto gp$, so it is smooth. Applying

the cited theorem gives

$$S^n \cong \mathrm{SO}(n+1)/\mathrm{SO}(n) \qquad (n \geq 1).$$

The identification is a diffeomorphism. In particular, the orbit map $g \mapsto ge_3$ exhibits $S^2 = \mathrm{SO}(3)/\mathrm{SO}(2)$. The subgroup $\mathrm{SO}(2)$ rotates the other two columns while leaving the last one fixed. These quotient descriptions concern smooth manifolds; the stabiliser is not a normal subgroup in general, so the quotient need not be a group.

For the qubit, use the left action $g \cdot [a : b] = [g(a,b)^T]$ of $\mathrm{SU}(2)$ on $\mathbb{C}\mathrm{P}^1$. The notation means the line spanned by the image column. If $|a|^2 + |b|^2 = 1$, the matrix with that first column is

$$U(a,b) = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}.$$

Its columns are orthonormal and its determinant is one, so $U(a,b) \in \mathrm{SU}(2)$. Conversely, a unitary matrix with first column $(a,b)^T$ has second column $\lambda(-\bar{b},\bar{a})^T$ for some $|\lambda| = 1$. Its determinant is λ, so determinant one forces $\lambda = 1$. Thus every matrix in $\mathrm{SU}(2)$ has this form. Transporting the smooth structure of S^3 through U makes $\mathrm{SU}(2)$ a smooth manifold. Its product and inverse are smooth: the first columns of the corresponding matrix formulas are polynomial in the real and imaginary coordinates. This makes $\mathrm{SU}(2)$ a Lie group. Every ray has a unit representative, proving transitivity. Fixing $[1 : 0]$ requires the first column to be $(e^{it}, 0)^T$. Unitarity and determinant one then force the other diagonal entry. Thus the stabiliser is the subgroup

$$H = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{-it} \end{pmatrix} : t \in \mathbb{R} \right\} \cong \mathrm{U}(1).$$

The action is smooth in projective charts: each output ratio is a ratio of complex linear expressions, on the open set where its denominator is nonzero. Consequently the cited theorem applies and gives $\mathbb{C}\mathrm{P}^1 \cong \mathrm{SU}(2)/\mathrm{U}(1)$ as smooth manifolds.

The figure places the two orbit maps and their stabilisers side by side.

$$\begin{array}{ccc} \mathrm{SO}(2) & \longrightarrow & \mathrm{SO}(3) \\ & & \downarrow{\scriptstyle g \mapsto ge_3} \\ & & S^2 \end{array} \qquad\qquad \begin{array}{ccc} \mathrm{U}(1) & \longrightarrow & \mathrm{SU}(2) \\ & & \downarrow{\scriptstyle g \mapsto [ge_1]} \\ & & \mathbb{C}\mathrm{P}^1 \end{array}$$

Figure 8.3: Each orbit map retains one datum: the last unit column for a rotation, or the line of the first unit column for $\mathrm{SU}(2)$. Multiplication on the right by its stabiliser leaves that datum fixed.

Example 8.6 (The Hopf map · *proved given the homogeneous-space theorem*)**.** The first-column map identifies $\mathrm{SU}(2)$ with $S^3 = \{(a,b) \in \mathbb{C}^2 : |a|^2 + |b|^2 = 1\}$. The inverse is $U(a,b)$, so both maps are smooth. The projection to the first-column line is the *Hopf map*

$$S^3 \longrightarrow \mathbb{C}\mathrm{P}^1, \qquad (a,b) \longmapsto [a : b].$$

Its fibre consists of the unit representatives $(e^{it}a, e^{it}b)$ of the same ray. Indeed, two unit columns span the same line exactly when the scalar relating them has modulus one. The matrix identity $U(e^{it}a, e^{it}b) = U(a,b)\,\mathrm{diag}(e^{it}, e^{-it})$ shows that these are precisely the left cosets $U(a,b)H$, the fibres of the orbit map $g \mapsto [ge_1]$. Composing with the Bloch diffeomorphism of Example 7.4 gives $S^3 \to S^2$. It forgets the common phase of a normalised qubit column. Lecture 20 develops its structure as a circle bundle.

Self-check. Find the dimension of $\mathrm{SO}(3)/\mathrm{SO}(2)$ using its stabiliser, and compare it with the tangent-plane calculation for the sphere.

Answer. The matrices in $\mathfrak{so}(3)$ have three independent parameters. The group $\mathrm{SO}(2)$ is a circle, so it has dimension one. The quotient dimension is $3 - 1 = 2$. The plane $p \cdot v = 0$ in $\mathbb{R}^3$ also has dimension two, since $p \neq 0$. The two calculations agree.

Summary. Tangent vectors are velocity classes. A chart identifies them with coordinate columns, and its Jacobian gives their transformation law. Differentials send curve velocities to image velocities and satisfy the chain rule. The tangent space of a regular level set is the kernel of the constraint's derivative. Closed stabilisers give smooth quotient manifolds by cited theorems. The rotation and qubit actions identify the sphere with $\mathrm{SO}(3)/\mathrm{SO}(2)$ and $\mathrm{SU}(2)/\mathrm{U}(1)$ respectively. Lecture 9 studies curves whose velocities are prescribed at every point by a vector field.

Reading. Tu, *An Introduction to Manifolds*, §§8, 12, 15–16; Lee, *Introduction to Smooth Manifolds*, second edition, Theorems 20.12, 21.17 and 21.18.